

Week 3

Annuity and Perpetuity

This week we will:

3.1: Distinguish the difference between an annuity and perpetuity -SLO2

3.2: Learn how to do annuity calculations (FV, PV and PMT) -SLO2

3.2a: Learn how to use a Financial Calculator to calculate FV of an annuity

3.2b: Learn how to use a Financial Calculator to calculate PV of an annuity

3.2c: Learn how to use a Financial Calculator to calculate PMT of an annuity

3.3: Learn how to do perpetuity calculation (PV and PMT) -SLO2

3.4: Put all different types of cashflows into the same timeline and calculate its Net Present Value - Net present value 2 - SLO4

Learning objectives:

- At the end of this module you will:

SLO2: apply time-value of money techniques to the valuation of financial instruments and financial securities

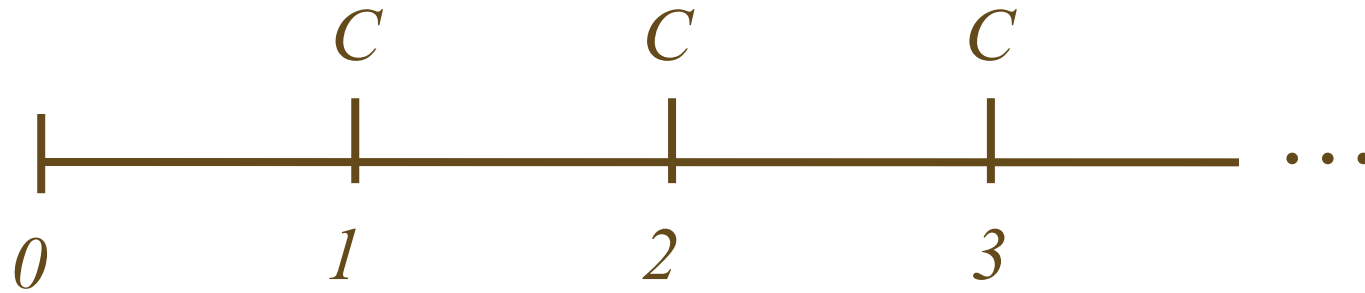
SLO4: evaluate a corporate investment proposal and provide a recommendation to accept or reject it

Simplifications

- Perpetuity
 - A constant stream of cash flows that lasts forever
- Growing perpetuity
 - A stream of cash flows that grows at a constant rate forever
- Annuity
 - A stream of constant cash flows that lasts for a fixed number of periods
- Growing annuity
 - A stream of cash flows that grows at a constant rate for a fixed number of periods

Perpetuity

A constant stream of cash flows that lasts forever



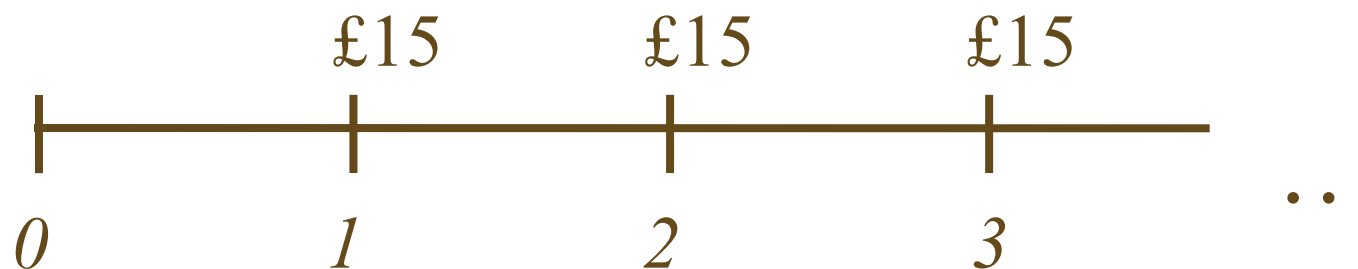
$$PV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \dots$$

$$PV = \frac{C}{r}$$

Perpetuity: Example

What is the value of a British consol that promises to pay £15 every year for ever?

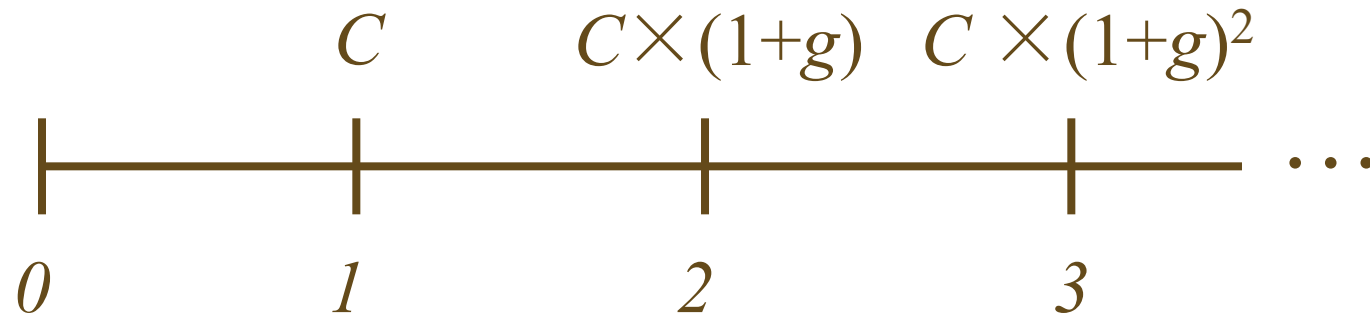
The interest rate is 10%.



$$PV = \text{£ } 15 / 0.10 = \text{£ } 150$$

Growing Perpetuity

A growing stream of cash flows that lasts forever



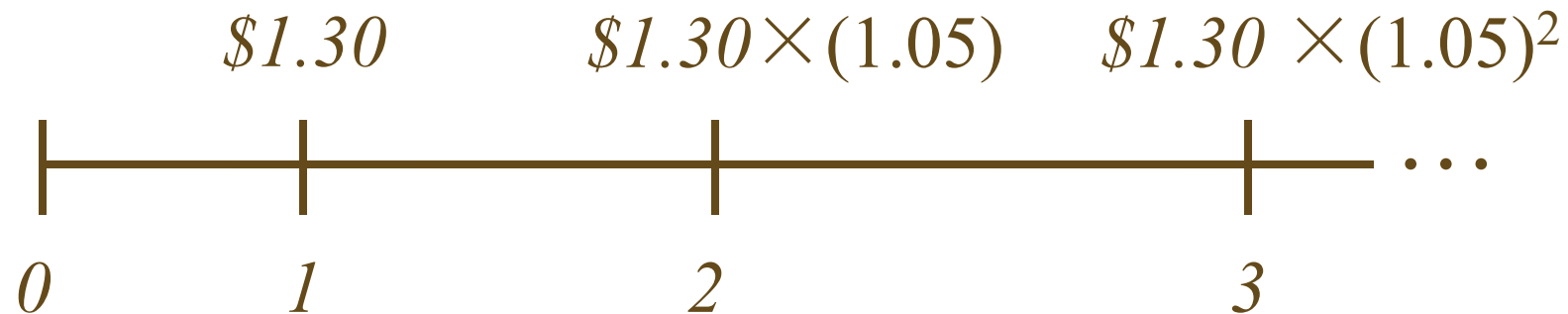
$$PV = \frac{C}{(1+r)} + \frac{C \times (1+g)}{(1+r)^2} + \frac{C \times (1+g)^2}{(1+r)^3} + \dots$$

$$PV = \frac{C}{r-g}$$

Growing Perpetuity: Example

The expected dividend next year is \$1.30, and dividends are expected to grow at 5% forever.

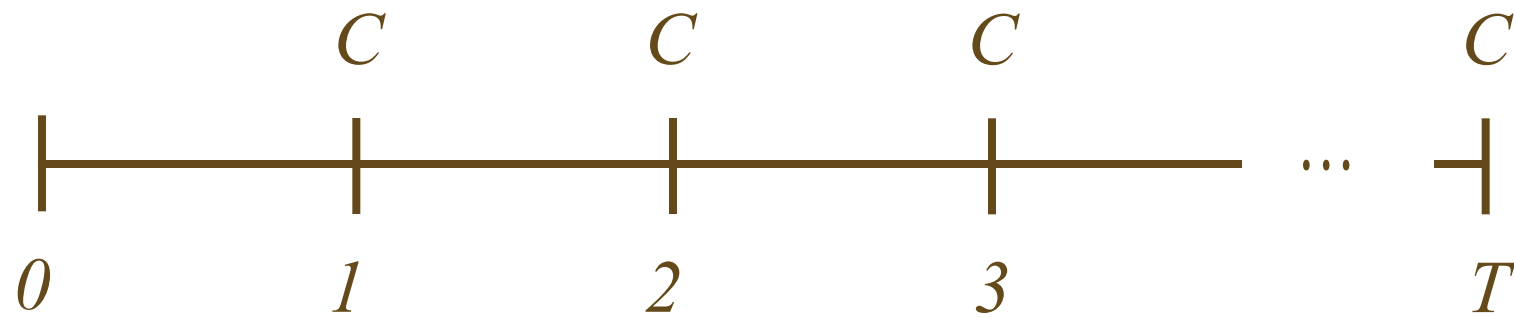
If the discount rate is 10%, what is the value of this promised dividend stream?



$$PV = \frac{\$1.30}{.10 - .05} = \$26.00$$

Annuity

A constant stream of cash flows with a fixed maturity

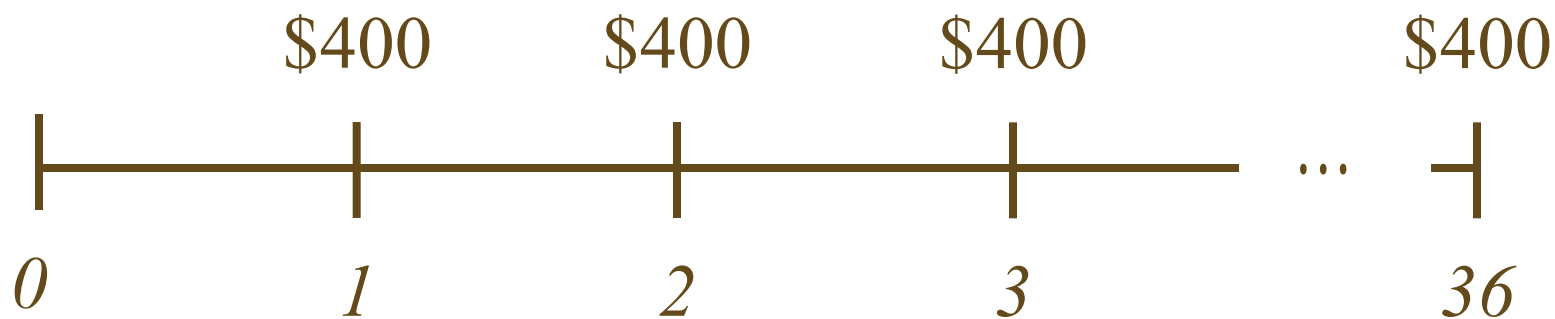


$$PV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \dots + \frac{C}{(1+r)^T}$$

$$PV = \frac{C}{r} \left[1 - \frac{1}{(1+r)^T} \right]$$

Annuity: Example

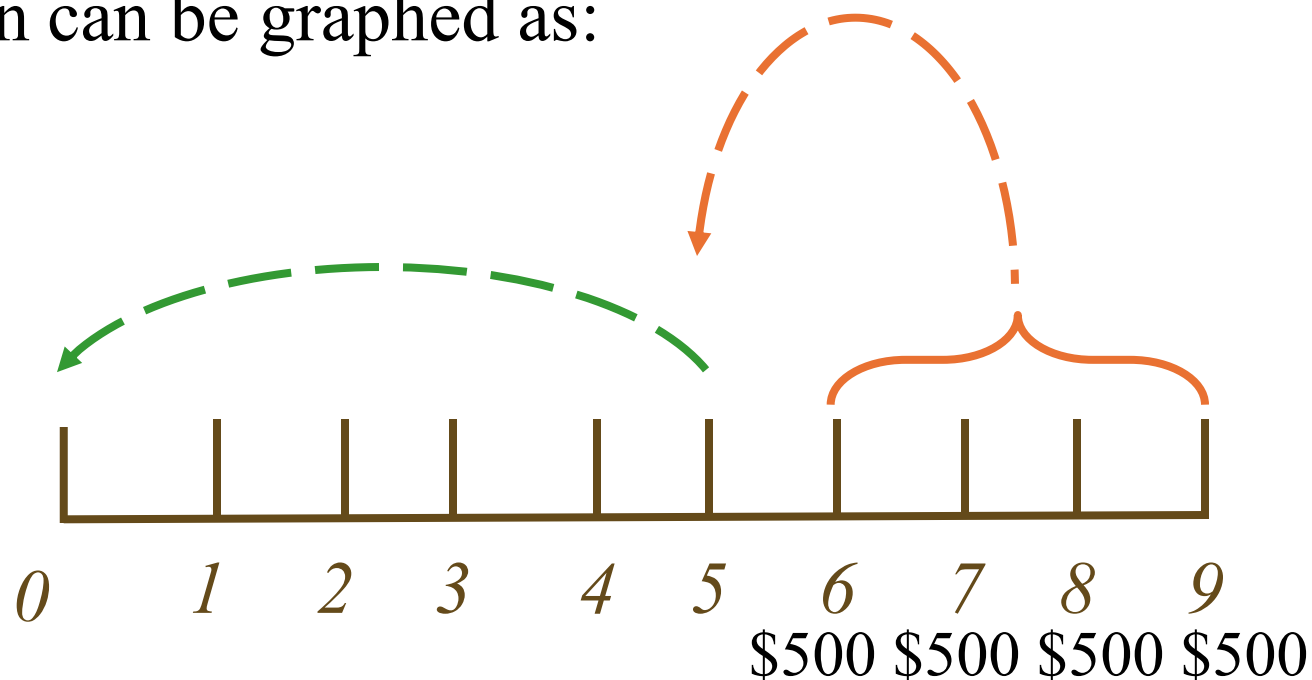
If you can afford a \$400 monthly car payment, how much car can you afford if interest rates are 7% on 36-month loans?



$$PV = \frac{\$400}{.07/12} \left[1 - \frac{1}{(1 + .07/12)^{36}} \right] = \$12,954.59$$

TRICK 1: A DELAYED ANNUITY

- Danielle Caravello will receive a four-year annuity of \$500 per year, beginning at date 6. If the interest rate is 10 percent, what is the present value of her annuity? This situation can be graphed as:



The analysis involves two steps:

1. Calculate the present value of the annuity using Formula 4.15.

This is

Present Value of Annuity at Date 5:

$$= \frac{\$500}{0.1} \left[1 - \frac{1}{(1+0.1)^4} \right]$$

$$= \$500 \times \text{PVIFA}_{10\%,4}$$

$$= \$500 \times 3.1699$$

$$= \$1,584.95$$

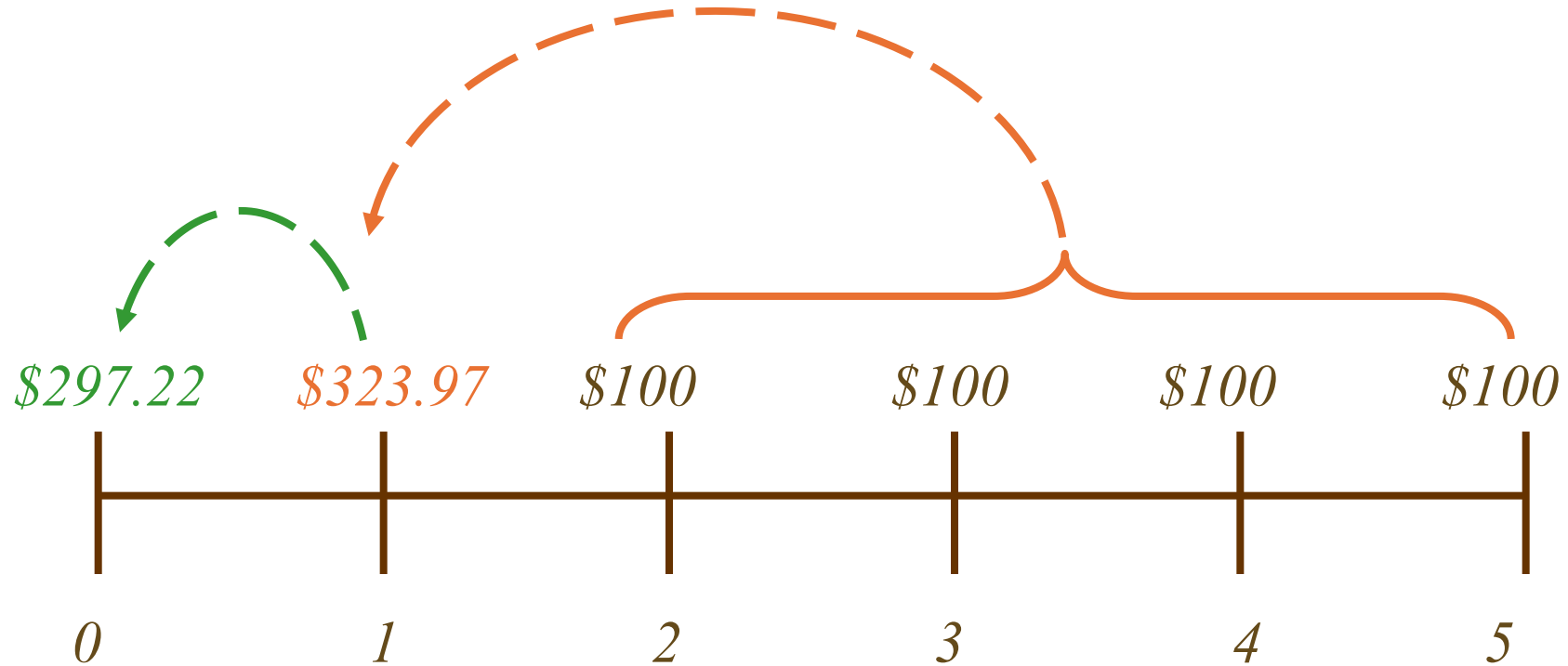
2. Discount the present value of the annuity back to date 0.

Present Value at Date 0:

$$= \$1,584.95 / (1.10)^5 = \$984.13$$

What is the present value of a four-year annuity of \$100 per year that makes its first payment two years from today if the discount rate is 9%?

$$PV_1 = \sum_{t=1}^4 \frac{\$100}{(1.09)^t} = \frac{\$100}{(1.09)^1} + \frac{\$100}{(1.09)^2} + \frac{\$100}{(1.09)^3} + \frac{\$100}{(1.09)^4} = \$323.97$$



$$PV_0 = \frac{\$327.97}{1.09} = \$297.22$$

Growing Perpetuities Mark Weinstein has been working on an advanced technology in laser eye surgery. His technology will be available in the near term. He anticipates his first annual cash flow from the technology to be \$210,000, received three years from today. Subsequent annual cash flows will grow at 2.5 percent in perpetuity. What is the value today of the technology if the discount rate is 11 percent?

答案： This is a growing perpetuity. The present value of a growing perpetuity is:

$$PV = C/(r - g)$$

$$PV = \$210,000/ (.11 - .025)$$

$$PV = \$2,470,588.24$$

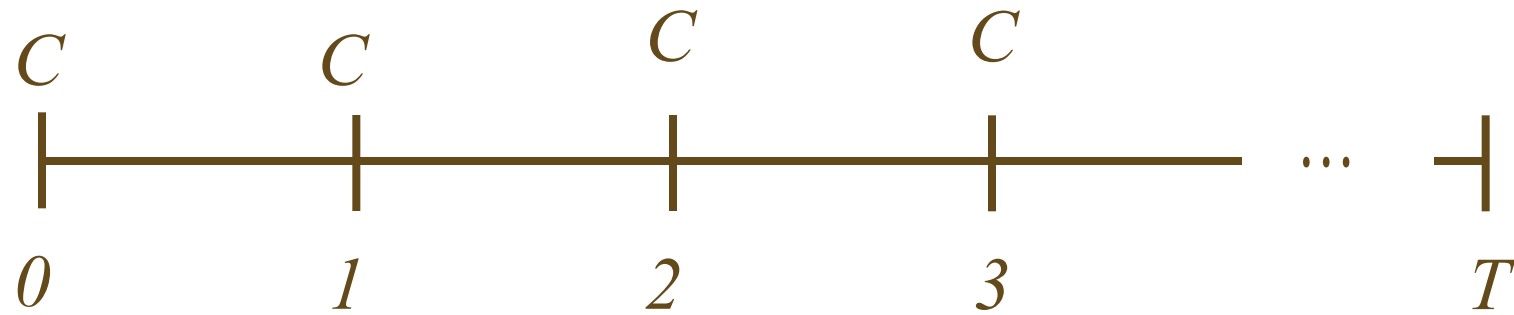
$$PV = FV/(1 + r)^t$$

$$PV = \$2,470,588.24/(1 + .11)^2$$

$$PV = \$2,005,184.84$$

TRICK 2: ANNUITY DUE

A constant stream of cash flows with a fixed maturity



$$PV = C + \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \dots + \frac{C}{(1+r)^{T-1}}$$

$$PV = C + \frac{C}{r} \left[1 - \frac{1}{(1+r)^{T-1}} \right]$$

Mark Young has just won the state lottery, paying \$50,000 a year for 20 years. He is to receive his first payment occurs. The state advertises this as the Million Dollar Lottery because $\$1,000,000 = \$50,000 \times 20$. If the interest rate is 8 percent, what is the true value of the lottery?

$$\begin{aligned} &\text{Present value of Million Dollar Lottery} \\ &= \$50,000 + \$50,000 \times \text{PVIFA}_{8\%,19} \\ &= \$50,000 + (\$50,000 \times 9.6036) \\ &= \$530,180 \end{aligned}$$

Calculating Present Values A 5-year annuity of 10 \$6,500 semiannual payments will begin 9 years from now, with the first payment coming 9.5 years from now. If the discount rate is 9 percent compounded monthly, what is the value of this annuity five years from now? What is the value three years from now? What is the current value of the annuity?

Answer: The cash flows in this problem are semiannual, so we need the effective semiannual rate. The interest rate given is the APR, so the monthly interest rate is:

$$\text{Monthly rate} = .09/12 = .0075$$

To get the semiannual interest rate, we can use the EAR equation, but instead of using 12 months as the exponent, we will use 6 months. The effective semiannual rate is:

$$\text{Semiannual rate} = (1.0075)^6 - 1 = 4.59\%$$

We can now use this rate to find the PV of the annuity. The PV of the annuity is:

$$\text{PVA @ } T = 9: \$6,500[(1 - 1/1.0459^{10})/.0459] = \$51,217.83$$

Note, that this is the value one period (six months) before the first payment, so it is the value at $t = 9$. So, the value at the various times the questions asked for uses this value 9 years from now.

$$\text{PV @ } T = 5: \$51,217.83/1.0459^8 = \$35,781.50$$

TRICK 3: THE INFREQUENT ANNUITY

Ms. Ann Chen receives an annuity of \$450, payable once every two years. The annuity stretches out over 20 years. The first payment occurs at date 2, that is, two years from today. The annual interest rate is 6 percent. What is the present value of the annuity?

The trick is to determine the interest rate over a two-year period. The interest rate over two years is

$$(1.06 \times 1.06) - 1 = 12.36\%$$

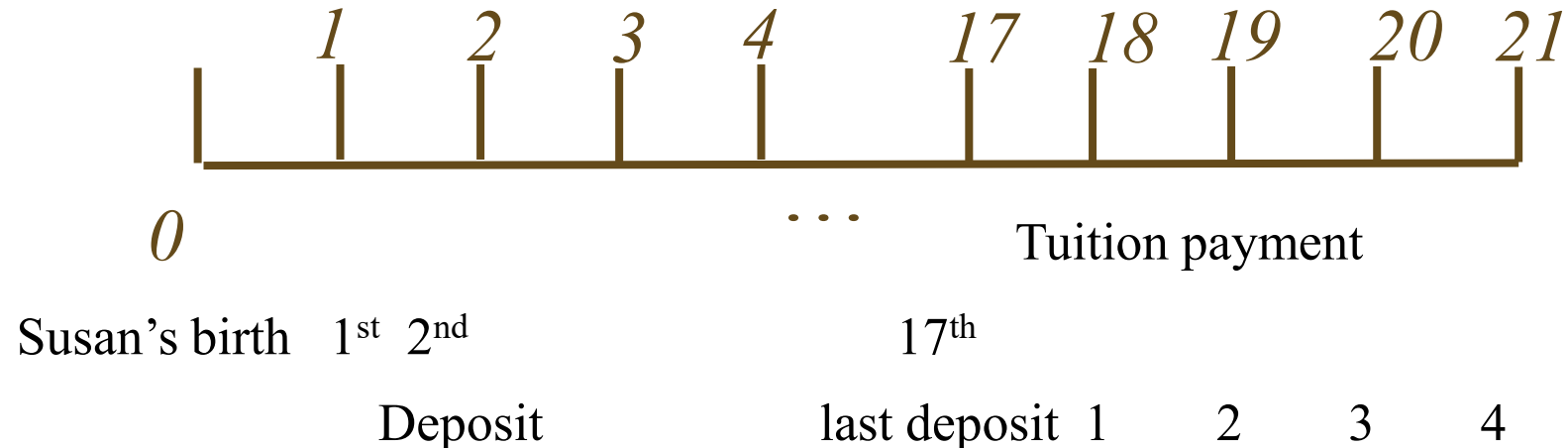
$$PV = \frac{\$450}{.01236} \left[1 - \frac{1}{(1 + 0.1236)^{10}} \right] = \$2,505.57$$

TRICK 4: equating present value of two annuities

Harold and Helen Nash are saving for the college education of their newborn daughter, Susan. The Nashes estimate that college expenses will run \$30,000 per year when their daughter reaches college in 18 years. The annual interest rate over the next few decades will be 14 percent. How much money must they deposit in the bank each year so that their daughter will be completely supported through four years of college?

TRICK 4: equating present value of two annuities

- Harold and Helen Nash are saving for the college education of their newborn daughter, Susan. The Nashes estimate that college expenses will run \$30,000 per year when their daughter reaches college in 18 years. The annual interest rate over the next few decades will be 14 percent. How much money must they deposit in the bank each year so that their daughter will be completely supported through four years of college?



1. We calculate the present value of the four years at college using the annuity formula.

$$\frac{\$30,000}{.014} \left[1 - \frac{1}{(1 + 0.14)^4} \right] = \$87,411$$

2. We calculate the present value of the college education at date 0 as

$$\frac{\$87,411}{1.14^{17}} = \$9,422.91$$

3. Assuming that Helen and Harold Nash make deposits to the bank at the end of each of the 17 years, we calculate the annual deposit that will yield a present value of all deposits of \$9,422.91. This is calculated as

$$C \times \frac{1}{14\%} \left[1 - \frac{1}{(1 + 14\%)^{17}} \right] = \$9,422.91$$

$$C = \$1,478.59$$

Future value of annuity

The following provides a formula for the future value of an annuity:

$$FV = C \left[\frac{(1 + r)^T}{r} - \frac{1}{r} \right] = C \left[\frac{(1 + r)^T - 1}{r} \right]$$

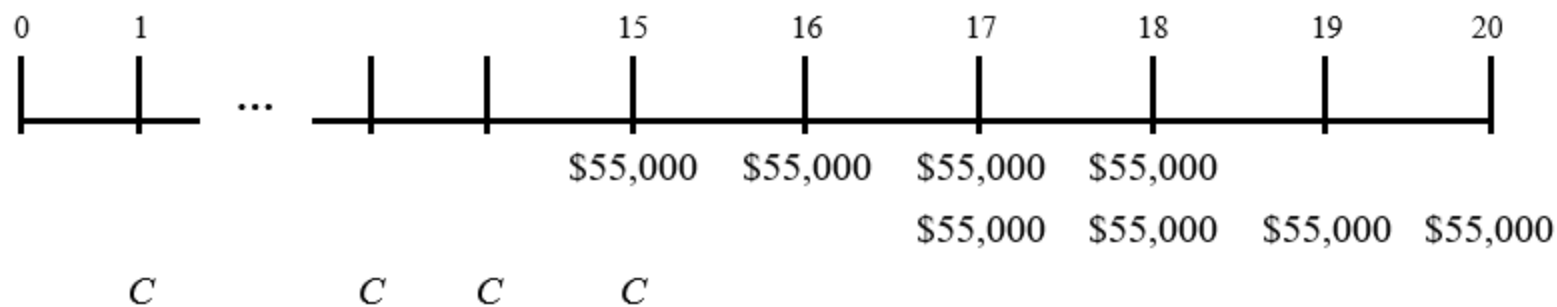
Suppose you put \$3,000 per year into a Roth IRA. The account pays 6 percent per year. How much will you have when you retire in 30 years?

This question asks for the future value of an annuity of \$3,000 per year for 30 years at 6 percent, which we can calculate as follows:

$$\begin{aligned} FV &= C \left[\frac{(1 + r)^T - 1}{r} \right] = \$3,000 \times \left[\frac{1.06^{30} - 1}{.06} \right] \\ &= \$3,000 \times 79.0582 \\ &= \$237,174.56 \end{aligned}$$

So, you'll have close to a quarter million dollars in the account.

Annuities You are saving for the college education of your two children. They are two years apart in age; one will begin college 15 years from today and the other will begin 17 years from today. You estimate your children's college expenses to be \$55,000 per year per child, payable at the beginning of each school year. The annual interest rate is 9.2 percent. How much money must you deposit in an account each year to fund your children's education? Your deposits begin one year from today. You will make your last deposit when your oldest child enters college. Assume your children will be on the four-year plan.



↵

$$PVA = C(\{1 - [1/(1 + r)]^t\}/r)$$

$$PVA = \$55,000(\{1 - [1/(1 + .092)^4]\}/.092)$$

$$PVA = \$177,405.18$$

$$PV = FV/(1 + r)^t$$

$$PV = \$177,405.18/(1 + .092)^{14}$$

$$PV = \$51,742.73$$

$$\text{Cost today} = \$51,742.73 + 43,391.44$$

$$\text{Cost today} = \$95,134.17$$

$$PV = FV/(1 + r)^t$$

$$PV = \$177,405.18/(1 + .092)^{16}$$

$$PV = \$43,391.44$$

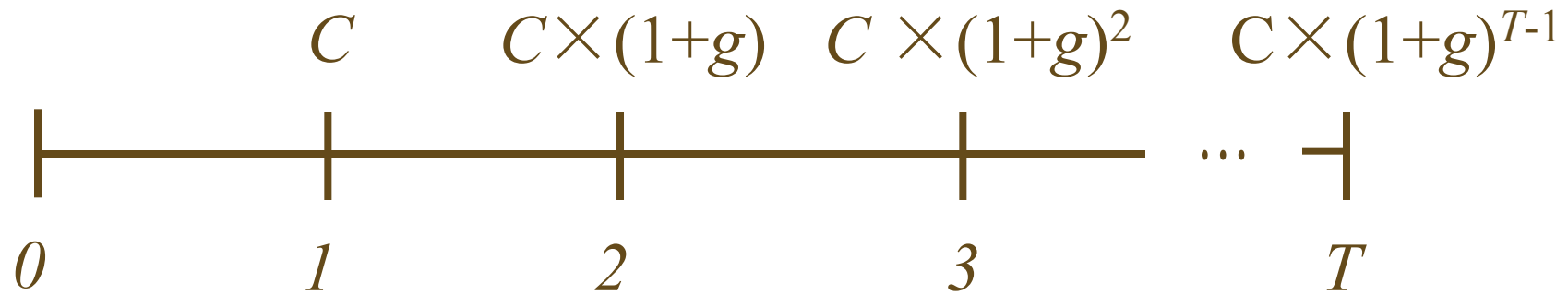
$$PVA = C(\{1 - [1/(1 + r)]^t\}/r)$$

$$\$95,134.17 = C(\{1 - [1/(1 + .092)^{15}]\}/.092)$$

$$C = \$11,941.94$$

Growing Annuity

A growing stream of cash flows with a fixed maturity

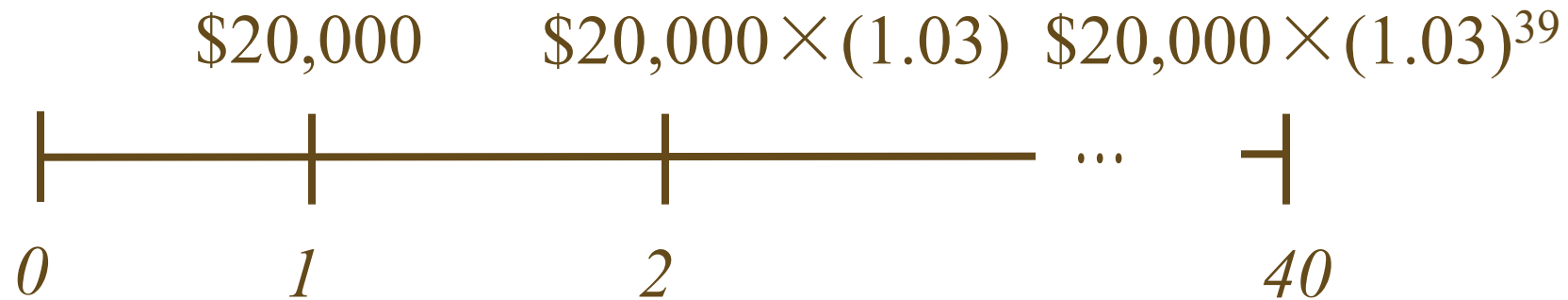


$$PV = \frac{C}{(1+r)} + \frac{C \times (1+g)}{(1+r)^2} + \dots + \frac{C \times (1+g)^{T-1}}{(1+r)^T}$$

$$PV = \frac{C}{r-g} \left[1 - \left(\frac{(1+g)}{(1+r)} \right)^T \right]$$

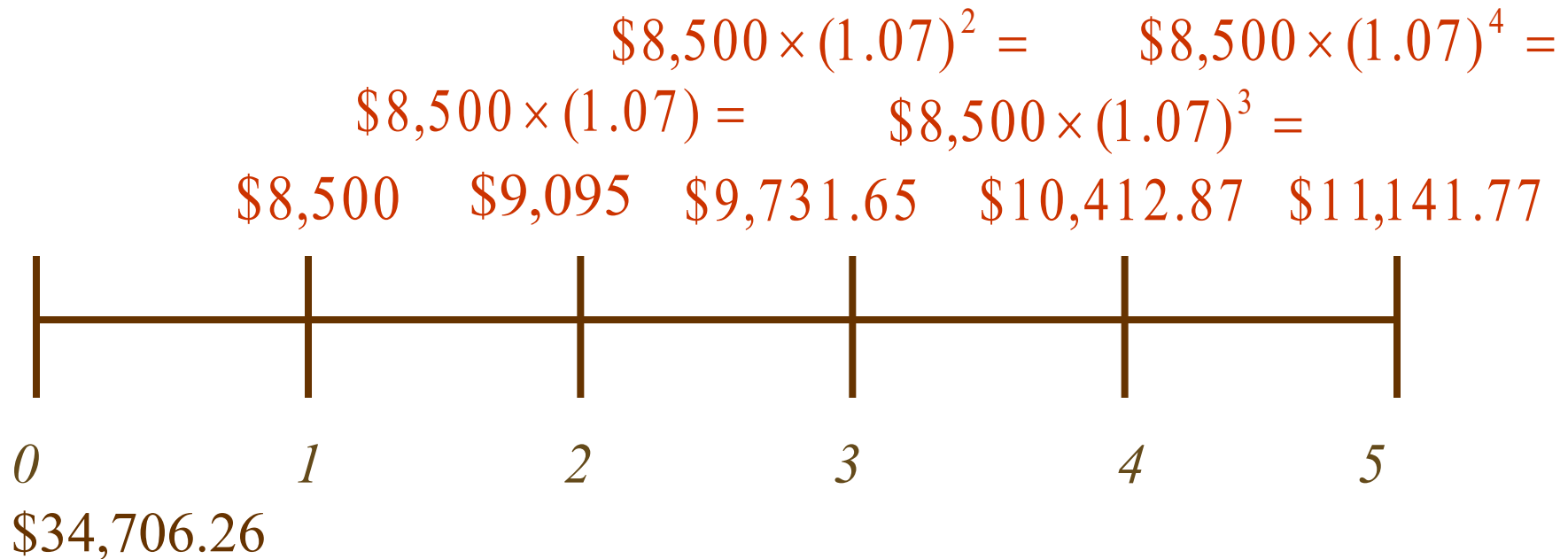
Growing Annuity: Example

A defined-benefit retirement plan offers to pay \$20,000 per year for 40 years and increase the annual payment by three-percent each year. What is the present value at retirement if the discount rate is 10 percent?



$$PV = \frac{\$20,000}{.10 - .03} \left[1 - \left(\frac{1.03}{1.10} \right)^{40} \right] = \$265,121.57$$

You are evaluating an income generating property. Net rent is received at the end of each year. The first year's rent is expected to be \$8,500, and rent is expected to increase 7% each year. What is the present value of the estimated income stream over the first 5 years if the discount rate is 12%?



Calculating Annuity Present Value An investment offers \$5,500 per year for 15 years, with the first payment occurring one year from now. If the required return is 7.5 percent, what is the value of the investment? What would the value be if the payments occurred for 40 years? For 75 years? Forever?

What Is a Firm Worth?

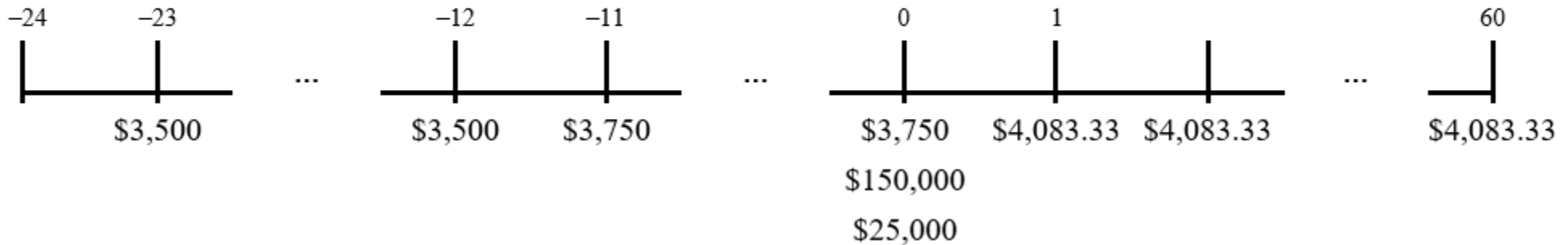
- Conceptually, a firm should be worth the present value of the firm's cash flows.
- The tricky part is determining the size, timing and *risk* of those cash flows.

A Note About Technique

- You can solve time value problems in any of four ways:
 - Math (Formulae given above)
 - Tables (See Appendix A)
 - Financial Calculator
 - Spreadsheet Software
- Financial calculators and spreadsheet software are the most common methods now.

Calculating Annuity Values You are serving on a jury. A plaintiff is suing the city for injuries sustained after a freak street sweeper accident. In the trial, doctors testified that it will be five years before the plaintiff is able to return to work. The jury has already decided in favor of the plaintiff. You are the foreperson of the jury and propose that the jury give the plaintiff an award to cover the following: (1) The present value of two years' back pay. The plaintiff's annual salary for the last two years would have been \$42,000 and \$45,000, respectively. (2) The present value of five years' future salary. You assume the salary will be \$49,000 per year. (3) \$150,000 for pain and suffering. (4) \$25,000 for court costs. Assume that the salary payments are equal amounts paid at the end of each month. If the interest rate you choose is an EAR of 9 percent, what is the size of the settlement? If you were the plaintiff, would you like to see a higher or lower interest rate?

The time line is:



$$\text{APR} = 12[(1.09)^{1/12} - 1] = 8.65\%$$

$$\text{FV} = (\$42,000/12)[\{[1 + (.0865/12)]^{12} - 1\}/(.0865/12)](1 + .09) = \$47,639.05$$

$$\text{FVA} = (\$45,000/12)[\{[1 + (.0865/12)]^{12} - 1\}/(.0865/12)] = \$46,827.37$$

$$\text{PVA} = (\$49,000/12)\{[\{1 - \{1/[1 + (.0865/12)]^{12(5)}\}]/(.0865/12)\} = \$198,332.55$$

$$\text{Award} = \$47,639.05 + 46,827.37 + 198,332.55 + 150,000 + 25,000$$

$$\text{Award} = \$467,798.97$$

Future Value and Multiple Cash Flows An insurance company is offering a new policy to its customers. Typically, the policy is bought by a parent or grandparent for a child at the child's birth. The details of the policy are as follows: The purchaser (say, the parent) makes the following six payments to the insurance company:

First birthday:	\$750
Second birthday:	750
Third birthday:	850
Fourth birthday:	850
Fifth birthday:	950
Sixth birthday:	950

After the child's sixth birthday, no more payments are made. When the child reaches age 65, he or she receives \$500,000. If the relevant interest rate is 10 percent for the first six years and 8 percent for all subsequent years, is the policy worth buying?

$$FV_1 = \$750(1.10)^5 = \$1,207.88$$

$$FV_2 = \$750(1.10)^4 = \$1,098.08$$

$$FV_3 = \$850(1.10)^3 = \$1,131.35$$

$$FV_4 = \$850(1.10)^2 = \$1,028.50$$

$$FV_5 = \$950(1.10)^1 = \$1,045$$

$$\text{Value at year six} = \$1,207.88 + 1,098.08 + 1,131.35 + 1,028.50 + 1,045 + 950$$

$$\text{Value at year six} = \$6,460.81$$

Finding the FV of this lump sum at the child's 65th birthday:

$$FV = \$6,460.81(1.08)^{59}$$

$$FV = \$605,742.96$$

The policy is not worth buying; the future value of the policy is \$605,742.96, but the policy contract will pay off \$500,000. The premiums are worth \$105,742.96 more than the policy payoff.